<u>Project Briefing</u>

Goals and Impact of a Successful Project

The goal of this project is to design and build a human-centered + agentic concierge application that makes high-quality, concierge-style planning accessible to a broad range of users. A successful project will demonstrate that users can move from a natural-language prompt to a concrete, shareable plan with minimal effort and confusion.

A. We will evaluate success using technical and user-focused metrics. On the technical side, the system must meet the standards required for the iOS App Store, including secure user data management, reliable API usage, and responsive performance. From a human-centered perspective, success will be measured by how quickly users can generate an itinerary, how well the system follows constraints (such as dietary restrictions or accessibility needs), and whether users feel confident sharing/acting on the generated itinerary.
B. This project is important because effective planning is often fragmented. Existing solutions require users to manually coordinate across maps, reviews, group chats, and booking platforms, which affects users with limited time, unfamiliarity with an area, or non-standard needs. In the short term, Navvy aims to reduce this by supporting natural-language itinerary creation and basic agentic behaviors. In the long term, the goal is to enable fully automated planning and booking workflows that can scale to users worldwide.
C. In comparison to our competitors, we will be the first to allow for natural language input and agentic capabilities regarding map apps and itinerary creation
D. A successful project will deliver an integrated map interface, natural-language processing for itinerary generation, and the foundation for agentic capabilities such as bookings and reservations.
E. Here is a short list of some user groups: urban users coordinating plans with others, tourists unfamiliar withthe local context, groups of friends with large coordination issues, and users with disabilities requiring accessible options

Background Information

A. Several existing applications address individual components of Navvy's scope, but none provide an integrated, intent-driven planning experience. Mapping applications such as Google Maps and Apple Maps excel at navigation and location search, but require users to manually discover places. Review and discovery platforms like Yelp or Beli support exploration but lack a planning structure and social coordination. Reservation platforms such as OpenTable or Resy handle transactions but assume that users already know where and when they want to go. Navvy builds on the strengths of these tools while addressing their fragmentation by treating the plan itself as the primary artifact.
    a. We can leverage everything mentioned in a combined app + agentic behavior: the Google Maps api for directions between locations, the Yelp API for providing a wide database of locations, and both OpenTable and Resy for allowing our agents to book reservations

B. There is new technology we have to create or develop. It does require some extensive understanding of building Agents, but we have team members who are extremely familiar with this, with foundational knowledge and professional experience
C. Several challenges are anticipated. One major challenge is interpreting vague or underspecified user prompts in a way that produces useful plans. Users often omit key details such as timing, group size, or budget, and the system must infer or request missing information without overwhelming the user. Another challenge is ensuring that recommendations respect hard constraints, such as dietary restrictions or accessibility needs.
D. A key technical hurdle is balancing responsiveness with reliability. AI-generated plans must be produced quickly enough to feel interactive, while also ensuring that locations, hours, and recommendations are accurate and current. Additionally, integrating multiple external APIs introduces potential failure modes related to latency, rate limits, and incomplete data.

Team Members, Skills, and Organization

We have no particular structure to our team, as we know we can all contribute equally, and we have a lot of overlapping skills and knowledge. We also don't have assigned responsibilities as we have all worked together before, and will divide work as we move on through development.

Aaron Li
- Major: Computer Science Engineering
- Key strengths: data aggregation, computer graphics, low-level procedural programming (C and C++)
- Skills to improve on: web development (backend and APIs), databases, mobile app development

Aryaman Goenka
- Major: Computer Science Engineering
- Key strengths: API Development, DB + Auth management, Agentic frameworks
- Skills to improve on: UI development (Human-centered UIs), A/B testing for deployment

Krish Kapoor
- Major: Computer Science Engineering
- Key strengths: Coding logic, agentic system development, prompt engineering
- Skills to improve on: Understanding large systems, database management, AI tools

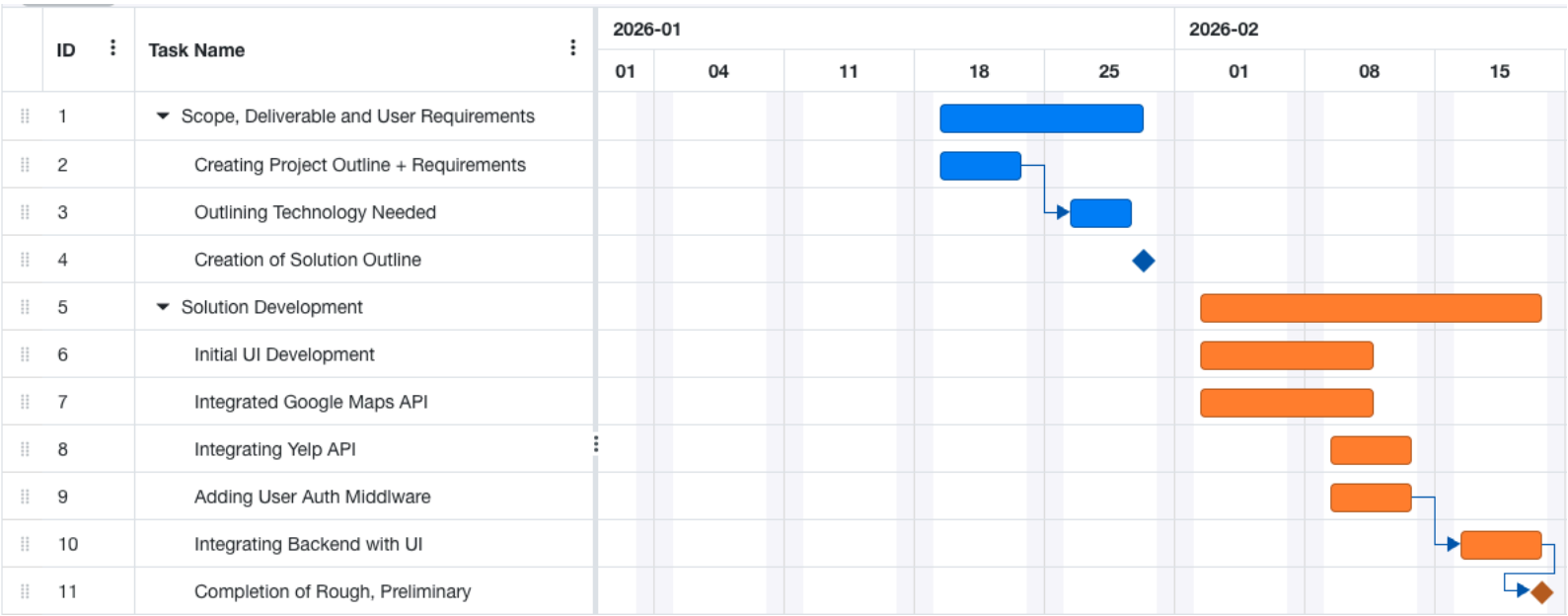
Project Management Strategy

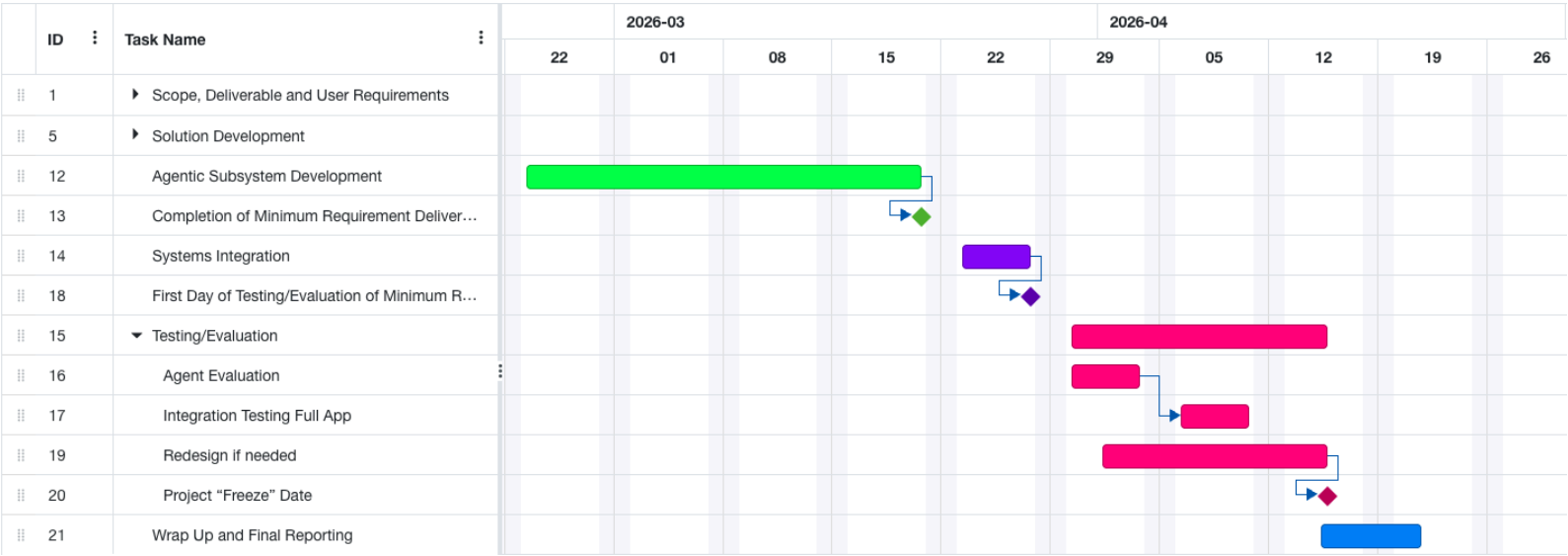
A. We will utilize GitHub as a remote to track our team's source code. We chose GitHub because it will keep our work organized and allow us to collaborate effectively. For our workflow, each member will work on a distinct feature, creating and pushing to their own feature branch as needed before merging their code into the main branch.
B. Each member in the team has had experience with Git, GitHub, and this workflow, and it has kept our work well organized in a team project setting.

C. We will adjust the expected timing/scope of the Milestones as needed, but we anticipate that we will not adjust them that often (perhaps 1-3 changes maximum throughout the entire semester). This project is one that we have been thinking about for a while, so the foundational plan, timeline of deliverables, and overall structure of the project are well understood among the group members.
D. Each group member has minimal experience with Project Management. We will need to add this skill to our list of new materials to learn.

Gantt Chart - Initial Project Management Plan & Milestones

Key: Rectangle = Phase, or Sub-Phase, Diamond = Milestone





Literature & Technology Review

Core Application Stack

- Frontend
    - React Native (Expo)
    - Map-based UI for spatial grounding and exploration
    - Emphasis on accessibility (dynamic type, contrast, screen reader support)
- Backend
  Fast-API Python backend layer
    - Orchestration layer for AI calls and third-party APIs
- Data Storage
    - Secure cloud database for user settings and saved plans on Supabase + Auth
    - Separation of personally identifiable information from generated content

Mapping & Location Services

- Google Maps / Places API
    - Core map rendering and geospatial context
    - Place metadata (hours, ratings, categories, location)
    - Distance and routing calculations
- Yelp Fusion API
    - Supplementary discovery and user reviews
    - Category-based filtering and popularity metrics
    - Accessibility Options
    - Used as a secondary signal rather than a source of truth

Reservations & Transactional Platforms (Future / Agentic Capabilities)

- Resy API + OpenTable API
    - High-end restaurant reservation access
    - Deposit-based and availability-aware booking
    - Critical for concierge-style experiences
- Viator API + TripAdvisor API
    - Event + Exercises + Activity reservation access

AI & Agent Tooling

- Large Language Models (LLMs)
- Agno (Agent Orchestration Framework)
    - Tool calling and action sequencing
    - Safe execution of agentic workflows (e.g., check availability → request confirmation → book)

While several existing products address parts of Navvy's functionality, they do not fully solve the underlying problem mentioned in this document. Tools such as Google Maps prioritize navigation and search, requiring users to manually discover activities. Review platforms like Yelp emphasize ratings and popularity but leave coordination entirely to the user. Reservation systems such as OpenTable assume that users already know where and when they want to go.

Navvy improves upon these approaches by focusing on intent-first interaction: users describe goals and constraints in natural language, and the system generates a coherent, editable itinerary that integrates discovery, logistics, and social coordination. This distinction is especially important for users with non-default needs—such as dietary restrictions, accessibility requirements, or large-group coordination—who are often underserved by existing tools.

At the AI layer, natural-language models are used to explicitly extract and classify user constraints (e.g., allergies, mobility limitations, budget ceilings) rather than treating them as optional filters. This ensures that constraints are applied before ranking or recommendation. Structured APIs provide metadata such as wheelchair accessibility, cuisine type, operating hours, and location density, allowing the system to verify that recommendations are compatible with user needs. Finally, agent tooling enforces confirmation and transparency before taking irreversible actions, such as making a paid reservation, which is especially important for users.

This AI layer will contain industry-standard LLMs for prompt parsing, intent extraction, and plan structuring. LLMs enable users to describe goals in their own words rather than conforming to rigid search interfaces. Agno's agent framework will allow multi-step, tool-based workflows such as checking availability, requesting user confirmation, and executing bookings. Agent tooling is intentionally constrained to ensure that actions affecting cost or commitment are transparent, confirmable, and reversible.

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